

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /Users/kaitlynkukula/EDE_Fall2025
```

```
library(agricolae)
library(ggplot2)
library(lubridate)
here()
```

```
## [1] "/Users/kaitlynkukula/EDE_Fall2025"
```

```
NTL_LTER_Raw <- read.csv("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv")
NTL_LTER_Raw$sampleddate <- mdy(NTL_LTER_Raw$sampleddate)
```

```
#2
```

```
kak_theme <- theme_classic() +
  theme(plot.title = element_text(size = 11, hjust = 0.5, face = "bold"), #Adjust title
        axis.text.x = element_text(size = 9), # Adjust x-axis values
        axis.text.y = element_text(size = 9), # Adjust y-axis values
        axis.title.x = element_text(size = 10.5, face = "bold"), # Adjust x-axis title
        axis.title.y = element_text(size = 10.5, face = "bold"), # Adjust y-axis title
        legend.position = "right", # Define legend position
        legend.text = element_text(size = 9), # Define legend entry sizes
        legend.title = element_text(size = 9)) # Define legend name sizes
set_theme(kak_theme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperature is not a function of depth in any lake Ha: Mean lake temperature is a function of depth in any lake
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
```

```
NTL_LTER_Processed <- NTL_LTER_Raw %>%
  filter(month(NTL_LTER_Raw$sampleddate) == "7") %>%
  select(c("lakename", "year4", "daynum", "depth", "temperature_C")) %>%
```

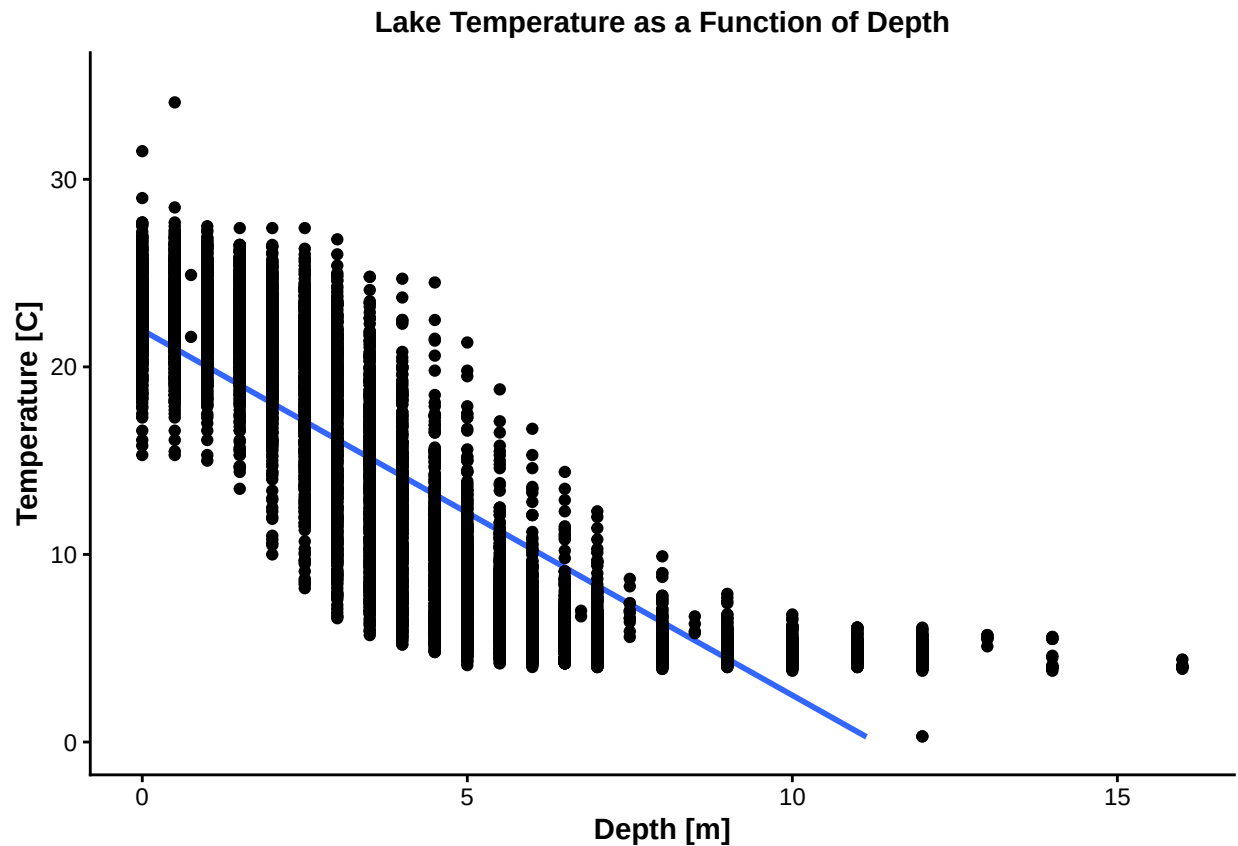
```
na.omit()

#5

NTL_LTER_plot <- ggplot(NTL_LTER_Processed, aes(depth, temperature_C)) +
  geom_smooth(method = "lm") +
  geom_point() +
  labs(x = "Depth [m]",
       y = "Temperature [C]",
       title = "Lake Temperature as a Function of Depth") +
  ylim(0, 35)
NTL_LTER_plot

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: This figure suggests that as depth increases, temperature decreases until a certain depth, where temperature becomes constant. It appears that the relationship between temperature and depth is not linear over the full depth analyzed. However, a linear relationship may exist for smaller depth ranges.

7. Perform a linear regression to test the relationship and display the results.

```
#7

NTL_LTER_LM <- lm(data = NTL_lTER_Processed, temperature_C ~ depth)
summary(NTL_LTER_LM)

##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL_lTER_Processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer:

This linear model produced the following line of best fit:

$$\text{Temperature} = -1.94 * \text{Depth} + 21.96$$

Based on this line, temperature decreases by 1.94 degC for every 1m change in depth. There were 9726 degrees of freedom, meaning that there were 9726 independent values in the calculation. The multiple R-squared and adjusted R-squared values were both 0.7387. This means that 73.87% of the response (temperature) can be explained by depth. The p-value for this model is 2.2e-16, which means that the data is significant and we should reject the null hypothesis in favor of the alternative hypothesis. Thus, based on this model, there is a significant relationship between depth and temperature.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

#9

```
NTL_AIC <- lm(data = NTL_1TER_Processed, temperature_C ~ year4 + daynum + depth)
step(NTL_AIC)
```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>             141687 26066
## - year4      1         101 141788 26070
## - daynum     1        1237 142924 26148
## - depth      1       404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_1TER_Processed)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##    -8.57556     0.01134     0.03978    -1.94644
```

#10

```
NTL_AIC_LM <- lm(data = NTL_1TER_Processed, temperature_C ~ year4 + daynum + depth)
summary(NTL_AIC_LM)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_1TER_Processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method suggests that all three of the tested variables should be used in a multiple regression. This new model explains 74.12% of the data, which is only a small improvement from the model that just considers depth.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
NTL_ANOVA <- aov(data = NTL_lTER_Processed, temperature_C ~ lakename)
summary(NTL_ANOVA)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2      50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
NTL_ANOVA_LM <- lm(data = NTL_lTER_Processed, temperature_C ~ lakename)
summary(NTL_ANOVA_LM)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTL_lTER_Processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769   -6.614   -2.679    7.684   23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake     -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake        -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake   -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

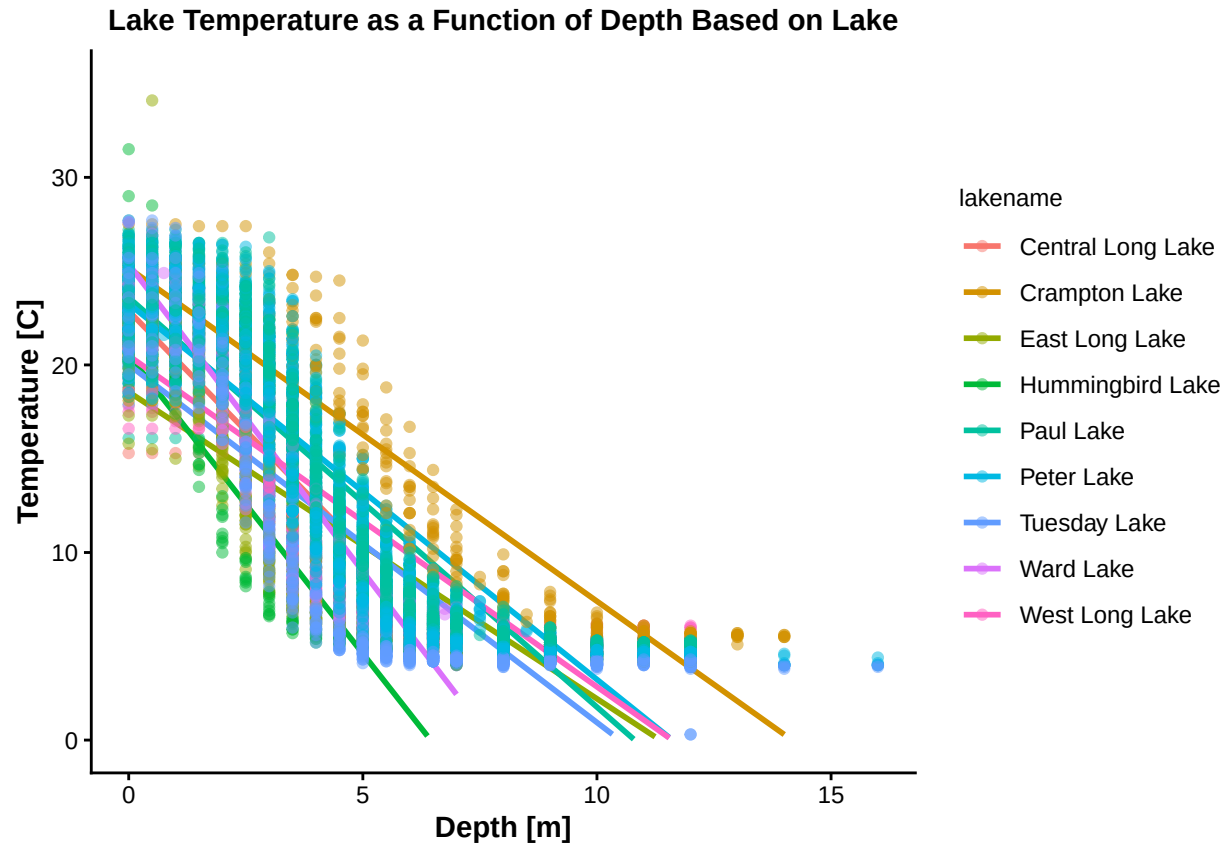
Answer: The ANOVA test resulted in a mean of 2705.2 with a p-value of 2e-16. Because the p value is < 0.05 , we are able to conclude that there is a significant difference in the temperatures among the lakes. The linear model resulted in a p value of 2.2e-16, so we can also conclude that there is a significant difference in the mean temperature among lakes. However, the linear model resulted in an R-squared value of 0.0395, meaning there is not a strong association between these variables.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
NTL_ANOVA_plot <- ggplot(NTL_lTER_Processed, aes(depth, temperature_C, color = lakename)) +
  geom_smooth(method = "lm", se = FALSE) +
  geom_point(alpha = 0.5) +
  ylim(0, 35) +
  labs(x = "Depth [m]",
       y = "Temperature [C]",
       title = "Lake Temperature as a Function of Depth Based on Lake")
print(NTL_ANOVA_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
NTL_HSD <- HSD.test(NTL_ANOVA, "lakename", group = TRUE)
NTL_HSD
```

```
## $statistics
##   MSerror  Df      Mean      CV
##   54.1016 9719 12.72087 57.82135
##
## $parameters
##   test  name.t ntr StudentizedRange alpha
##   Tukey lakename  9      4.387504  0.05
##
## $means
##               temperature_C      std      r      se Min  Max   Q25  Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake             13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake           11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
## Ward Lake              14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
```



```
## West Long Lake      11.57865 6.980789 1026 0.2296314 4.0 25.7 5.400 8.00
##                      Q75
## Central Long Lake 21.000
## Crampton Lake      22.300
## East Long Lake      15.925
## Hummingbird Lake    15.625
## Paul Lake           21.400
## Peter Lake          21.500
## Tuesday Lake        19.400
## Ward Lake           23.200
## West Long Lake      18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake           11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer:

The following lakes have the same mean temperatures:

Central Long Lake and Crampton Lake Crampton Lake and Ward Lake Ward Lake, Paul Lake, Peter Lake West Long Lake, Tuesday Lake, Hummingbird Lake Tuesday Lake, Hummingbird Lake, East Long Lake

There are no lakes that are statistically distinct from the others.

17. If we were just looking at Peter Lake and Paul Lake, what's another test we might explore to see whether they have distinct mean temperatures?

Answer: You could use a two-way T-test to determine if two variables (the two lakes) have the same mean.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```

NTL_Processed_Crampton_Ward <- NTL_1TER_Processed %>%
  filter(lakename == c("Crampton Lake", "Ward Lake"))

NTL_Crampton_Ward_ttest <- t.test(NTL_Processed_Crampton_Ward$temperature_C ~ NTL_Processed_Crampton_Ward$lakename)
NTL_Crampton_Ward_ttest

```

```

##
## Welch Two Sample t-test
##
## data: NTL_Processed_Crampton_Ward$temperature_C by NTL_Processed_Crampton_Ward$lakename
## t = 0.98673, df = 95.77, p-value = 0.3263
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -1.130614 3.365610
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.37107                14.25357

```

Answer: This test resulted in a mean value of 15.37 for Crampton Lake and 14.25 for Ward Lake. The p-value of this test is 0.3263, so we cannot reject the null hypothesis. However, the null hypothesis is that the true difference in the means is equal to zero, so these two lakes have the same mean temperature. This matches the answer found in part 16.